8.4.1 Petrol System

This type of lubricating system is used in small vehicles like in two wheelers (e.g. TVS 50 and other mopeds and also in Motor cycles with two stroke engines. In this system lubricating oil is of about 2% to 3% is mixed along with petrol and used. It is a simple lubricant system commonly used in two stroke two wheelers and small size engines.

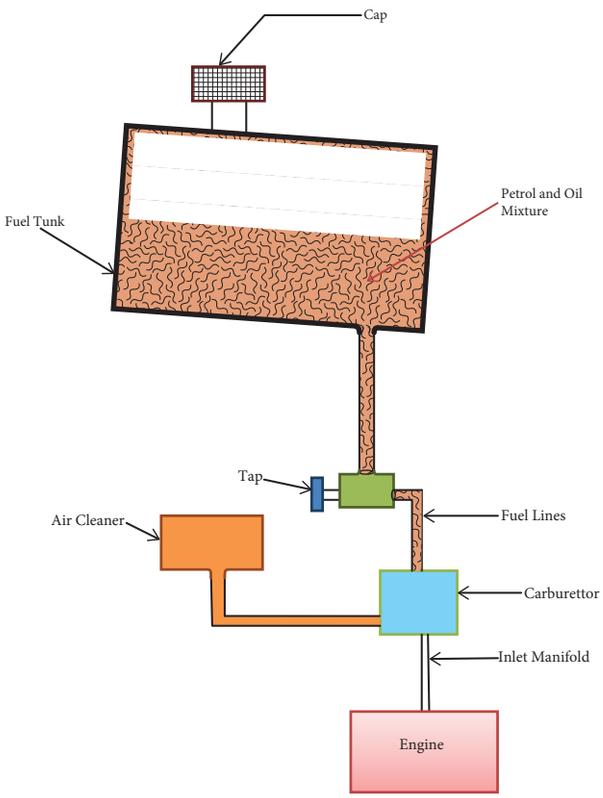


Figure 8.4.1 Petrol Oil Lubrication System

8.4.2 Splash Lubrication System

In this system, at the bottom of the crank case, oil sump will be present and filled with the lubricating oil. The engines with this type of system have dipper or scoop like arrangement fitted at the bottom of the connection road. When the piston moves towards bottom dead centre, the scoop placed at the connecting rod dips into the oil sump and scoops the oil. When the piston moves up the oil in the scoop will be splashed on all over the interior of the crankcase, into the piston and to the exposed portions of the cylinder wall and other components. See Figure shown in 8.4.2 Splash Lubrication System

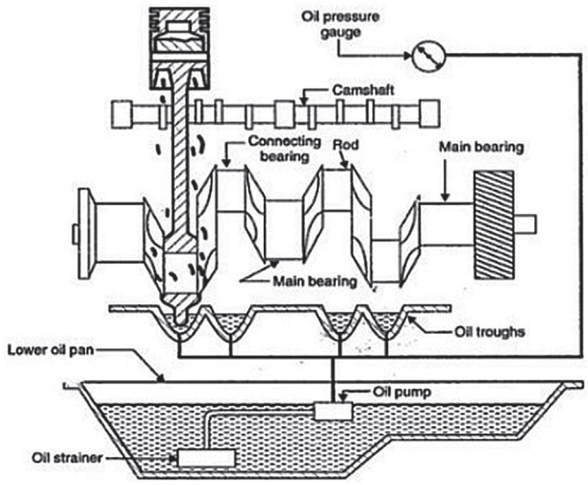


Figure 8.4.2 Splash Lubrication System

8.4.3 Pressure Lubrication

Sending the lubricating oil at high pressure using a pump to all the parts of engines is called pressure lubrication. In this system oil is kept at the base of crank case. During operation of the engine, oil pump sucks the oil from sump through the strainer. This oil is subjected to high pressure of about 200 kPa to 400 kPa and then sent to the oil filter. It is then filtered in the oil filter and sent to main gallery. From the main gallery the oil is sent to main journal bearings and sub journal bearings of the crank shaft for lubrication. The oil is then sent to the piston pin and piston rings via the oil hole of connecting

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